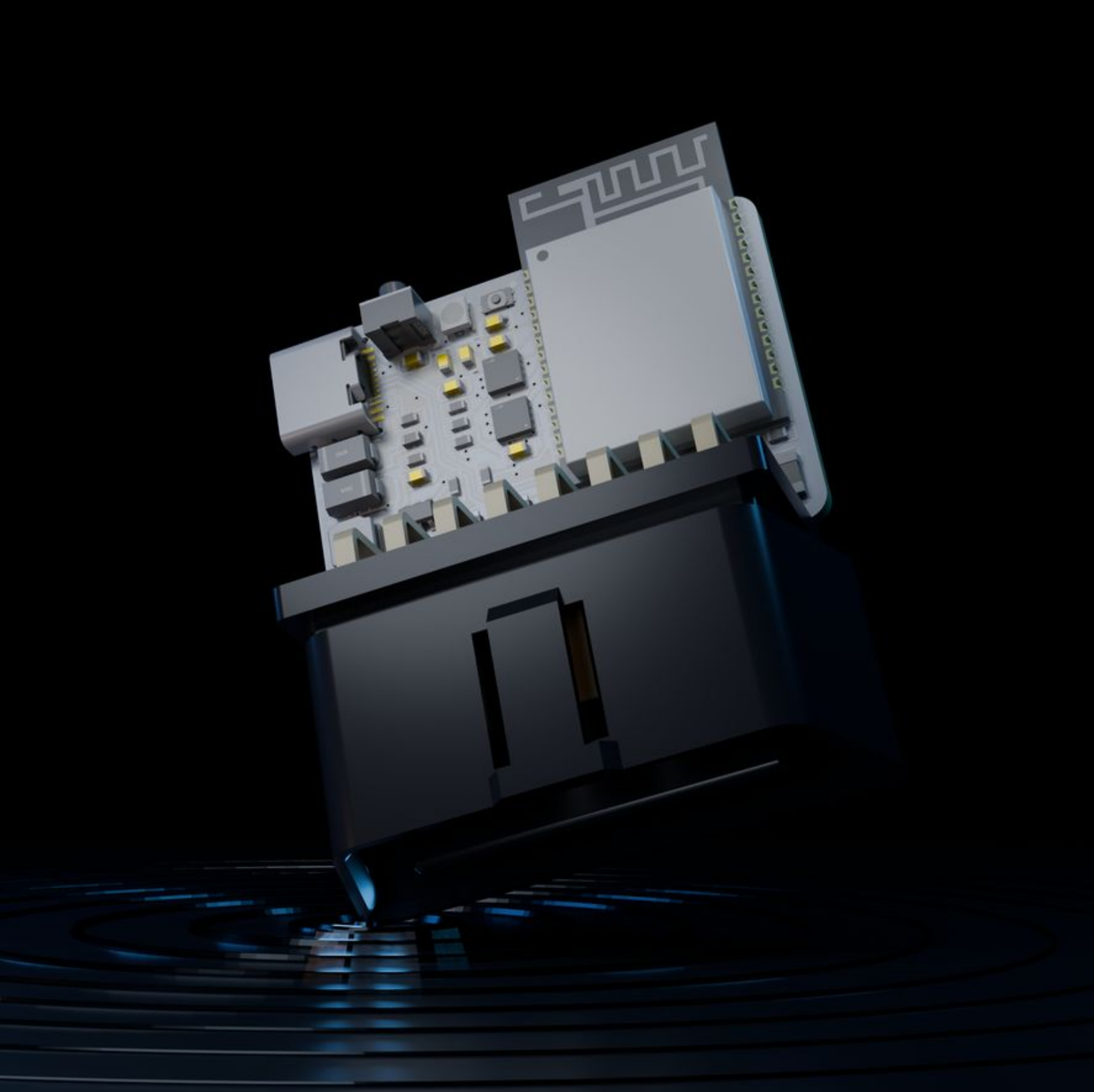




CANOpener SE Dual CAN Bus Vehicle Interface Hardware Documentation

**Standard Edition
Hardware Revision A**

INTRODUCTION ---

Canopener SE is a compact dual CAN bus interface designed to connect directly to vehicle networks through the OBD-II diagnostic port.

At the heart of Canopener SE is an Espressif ESP32-C5 microcontroller with two independent CAN-FD controllers and integrated wireless connectivity. Two Texas Instruments TCAN3414 CAN transceivers provide the physical interface between the controller and the vehicle CAN networks.

The device connects directly to two CAN buses available through the OBD-II connector and supports both Classical CAN and CAN FD communication. Canopener SE can be powered directly from the vehicle or through USB-C for development and configuration.

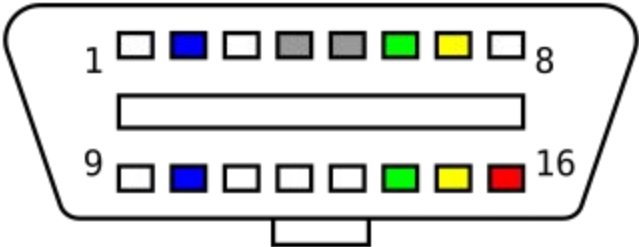
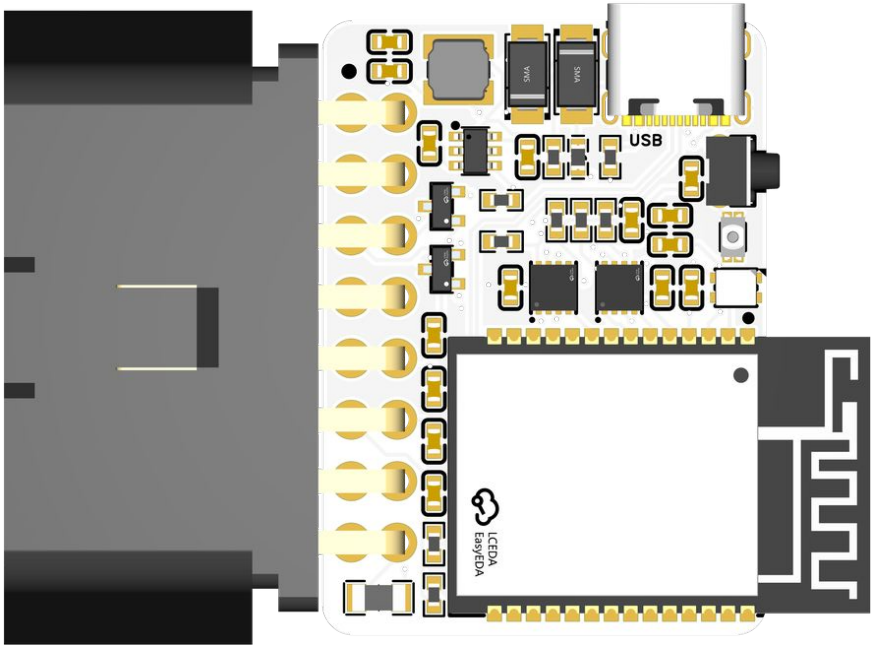
Onboard voltage regulation, input protection, CAN bus protection, an addressable RGB status LED, and dedicated Boot and Reset controls are integrated into the board.

ELECTRICAL SPECIFICATIONS ---

Vehicle Input Voltage	Up to 24 V
Typical Input Voltage	12 V
USB Input Voltage	5 V
Logic / I/O Voltage	3.3 V
Operating Current	-
CAN Interfaces	2
CAN Protocols	Classical CAN, CAN FD
Maximum CAN Data Rate	5 Mbps

Note: ...

HARDWARE OVERVIEW



1	NC	9	NC
2	NC	10	NC
3	CAN1 High	11	CAN1 Low
4	GND	12	NC
5	NC	13	NC
6	CAN0 High	14	CAN0 Low
7	NC	15	NC
8	NC	16	Vehicle Power

PHYSICAL SPECIFICATIONS

